

Lab 3: Simulation of Network Devices (Hub, Switch, Bridge, Router, Repeater) Using Cisco Packet Tracer

❖ Objective:

- To understand the working principles of basic network devices.
- To simulate hub, switch, bridge, router, and repeater using Cisco Packet Tracer.
- To observe data transmission and packet flow in different network devices.

❖ Theory:

Computer networking is the practice of connecting multiple computing devices to share data, resources, and services. Network devices are essential components that control how data is transmitted across a network.

A **hub** is a simple networking device that operates at the physical layer of the OSI model. It broadcasts incoming data packets to all connected devices, which can lead to network congestion and collisions.

Figure:



A **switch** operates at the data link layer and forwards data only to the intended recipient using MAC addresses. This improves network efficiency and reduces unnecessary traffic.

Figure:



A **bridge** is used to divide a network into smaller segments, reducing collisions and improving performance. It filters traffic based on MAC addresses and works similarly to a switch but with fewer ports.

Figure:



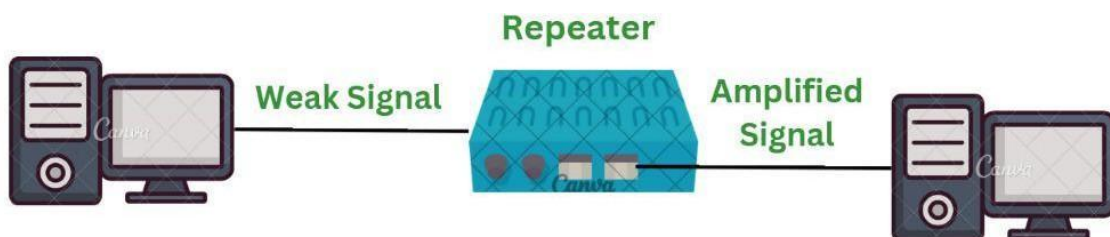
A **router** operates at the network layer and connects multiple networks together. It uses IP addresses to determine the best path for data packets, enabling communication between different networks.

Figure:



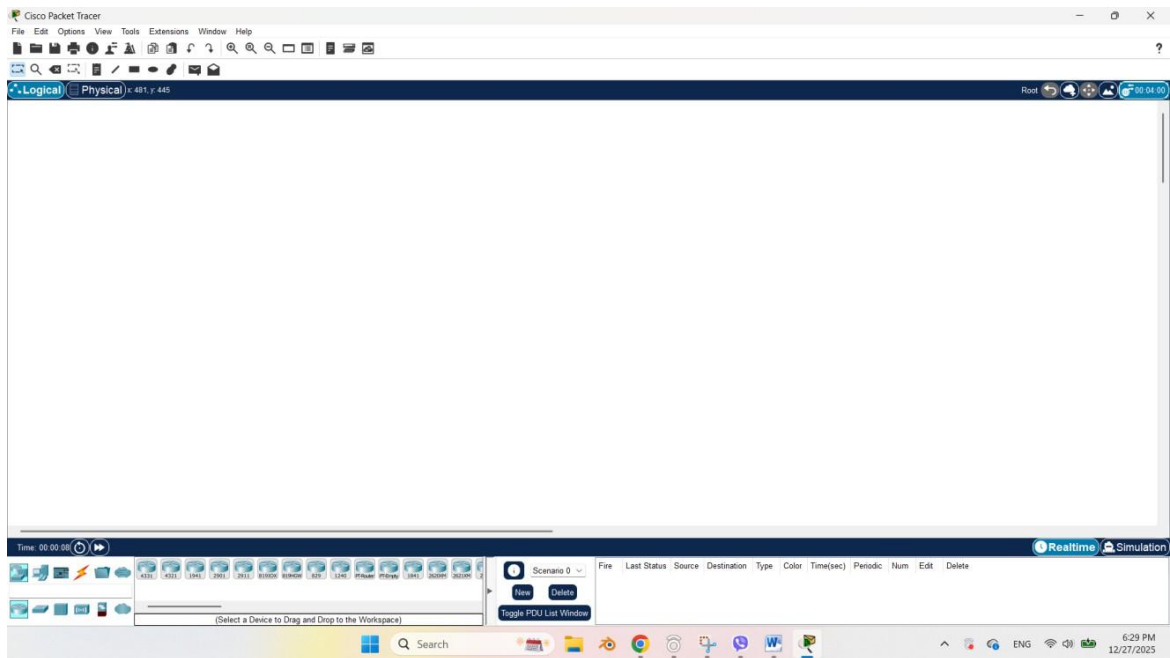
A **repeater** is a physical layer device that regenerates and amplifies signals to extend the distance of data transmission. It is commonly used in long network cables to prevent signal degradation.

Figure:



Cisco Packet Tracer is a network simulation tool developed by Cisco that allows users to design, configure, and test network topologies in a virtual environment. It helps students understand networking concepts without requiring physical hardware.

Figure:



❖ Working method with output:

1. Hub

Working Method:

- First, we open **Cisco Packet Tracer** on the system.
- Then, we select **end devices (PCs)** and place them on the workspace.
- We add a **hub** and connect it to multiple PCs using **straight-through cables**.
- Next, we send data packets from one PC, and we observe that the **hub broadcasts the data to all connected devices**.
- This demonstrates that a hub **does not differentiate between devices** and sends data to every port, making it suitable for small networks but less efficient for larger ones.

Ping from PC1 to PC3:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.4

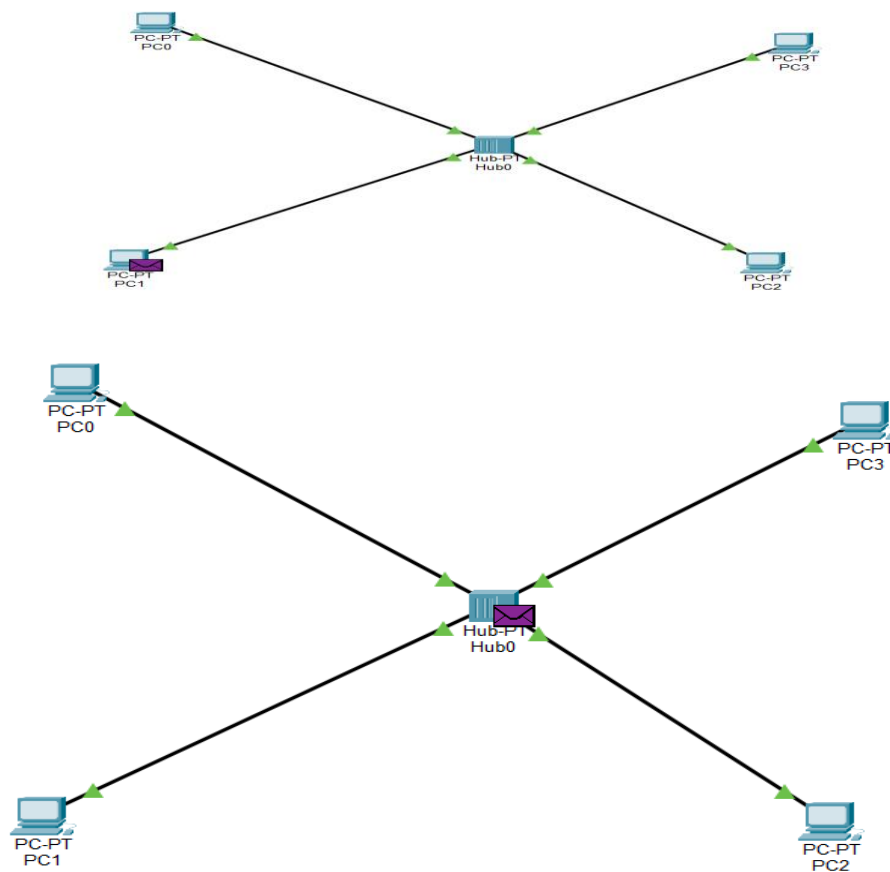
Pinging 192.168.1.4 with 32 bytes of data:

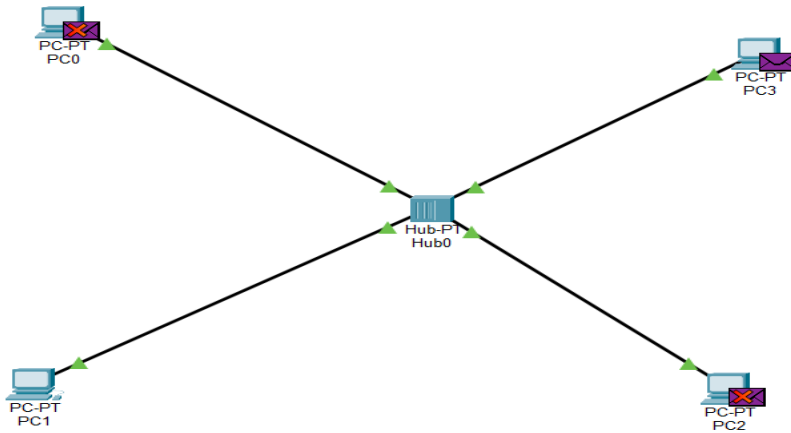
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=7ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 1ms

C:\>|
```

Output:





2. Switch

Working Method:

- After testing the hub, we **replace the hub with a switch** and reconnect the PCs using straight-through cables.
- We then send data packets from one PC and observe that the **switch forwards the data only to the intended destination device**.
- This happens because switches **maintain a MAC address table**, which allows them to **filter and direct traffic efficiently**.
- We also notice reduced network congestion compared to the hub, highlighting the **advantages of switches in larger networks**.

Ping from PC0 to PC1:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.6

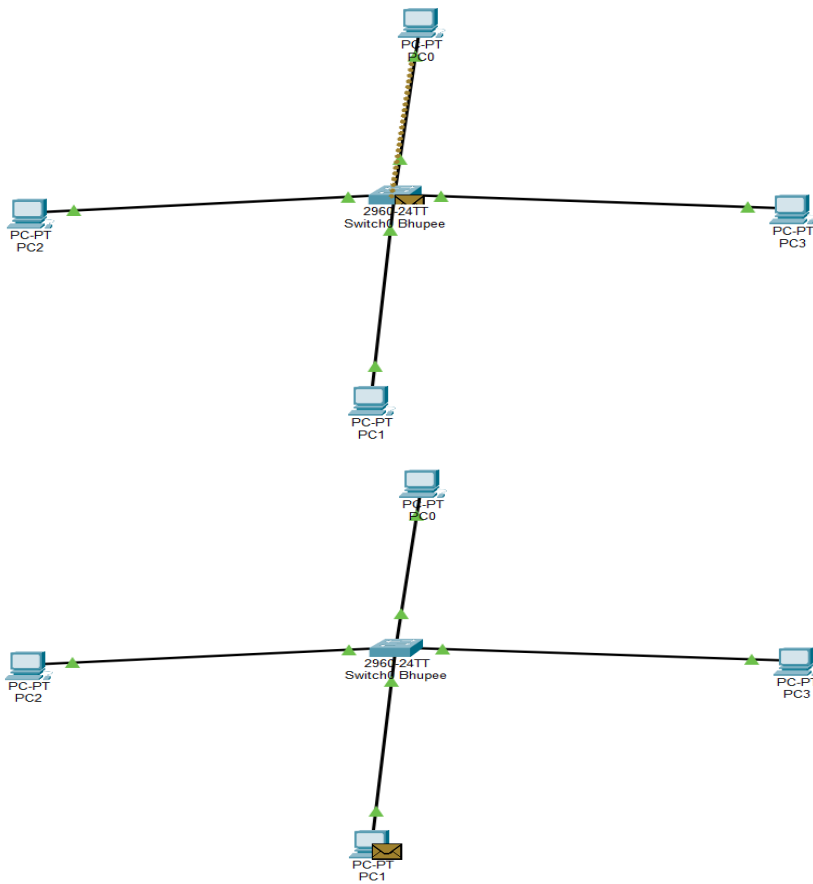
Pinging 192.168.1.6 with 32 bytes of data:

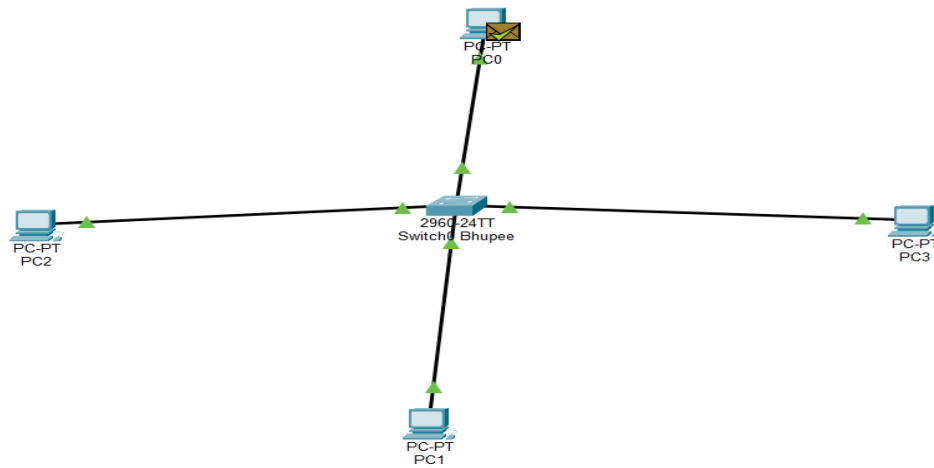
Reply from 192.168.1.6: bytes=32 time=1ms TTL=128
Reply from 192.168.1.6: bytes=32 time=1ms TTL=128
Reply from 192.168.1.6: bytes=32 time=10ms TTL=128
Reply from 192.168.1.6: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>
```

Output:





3. Bridge

Working Method:

- We insert a **bridge between two network segments** in the workspace.
- PCs and Laptops from both segments are connected to the bridge using appropriate cables.
- Then, we send data packets between devices from different segments and observe that the **bridge filters and forwards data based on MAC addresses**.
- The bridge helps **segregate network traffic**, which reduces unnecessary load and **enhances network efficiency**.

Ping form PC0 to PC4:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.6

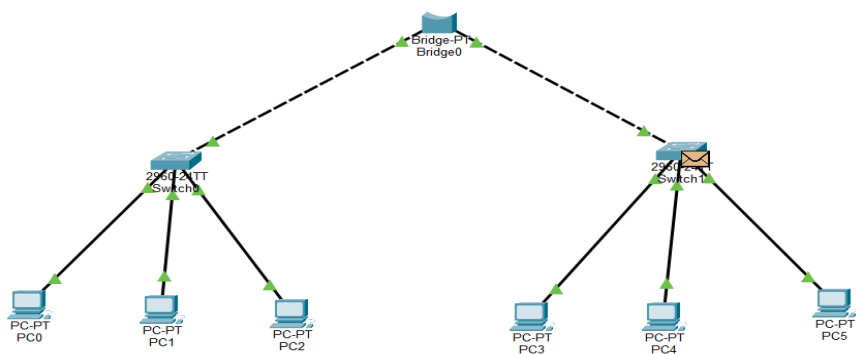
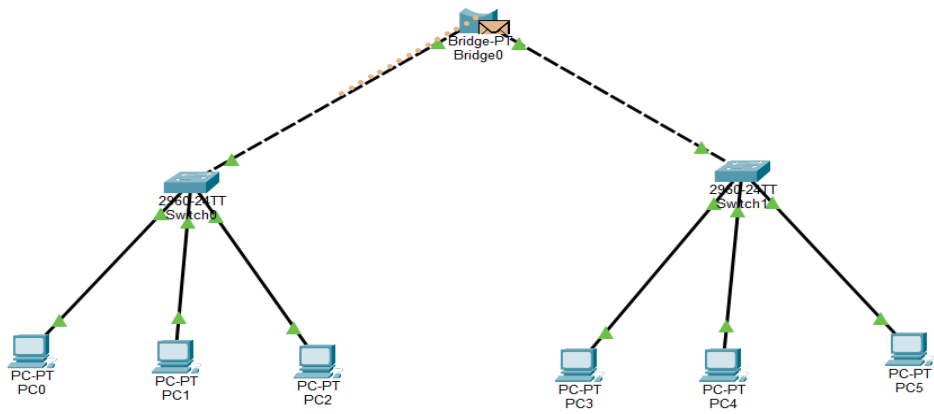
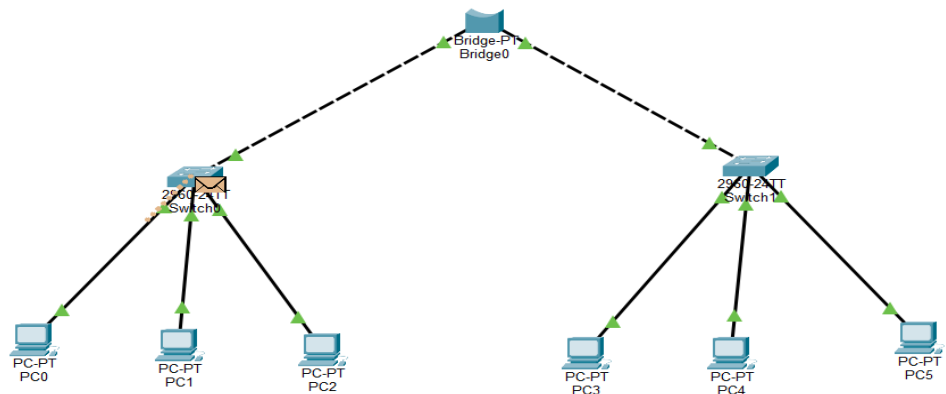
Pinging 192.168.1.6 with 32 bytes of data:

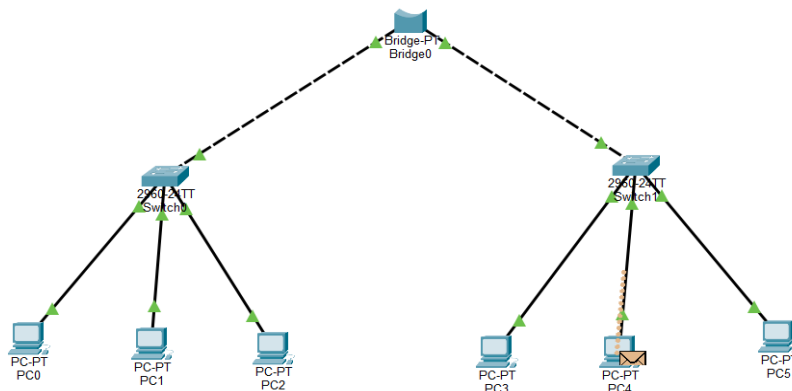
Reply from 192.168.1.6: bytes=32 time=1ms TTL=128
Reply from 192.168.1.6: bytes=32 time=1ms TTL=128
Reply from 192.168.1.6: bytes=32 time=10ms TTL=128
Reply from 192.168.1.6: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>
```

Output:





4. Router

Working Method:

- Next, we add a **router** to the workspace and connect it to two different networks.
- We configure **IP addresses** for each interface of the router to ensure proper routing.
- We then test connectivity using the **ping command** between devices in different networks.
- The router **routes data packets based on IP addresses**, allowing **communication between separate networks**.
- We observe successful data transfer, demonstrating how routers **enable inter-network communication** and manage traffic intelligently.

Ping from PC0to PC3:

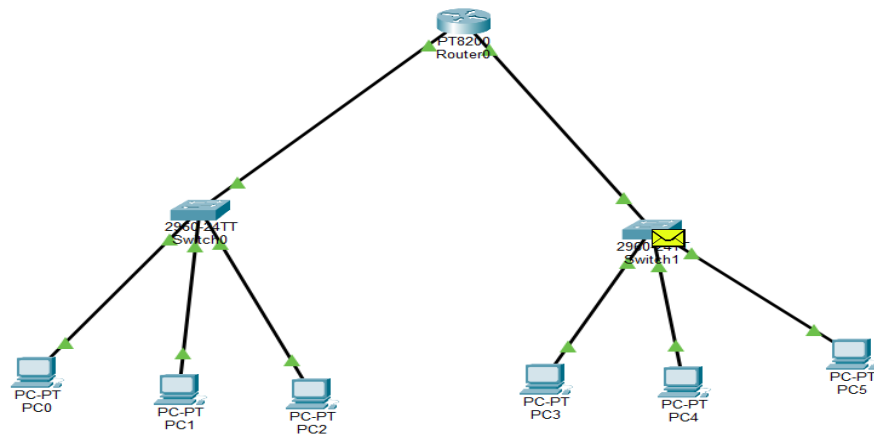
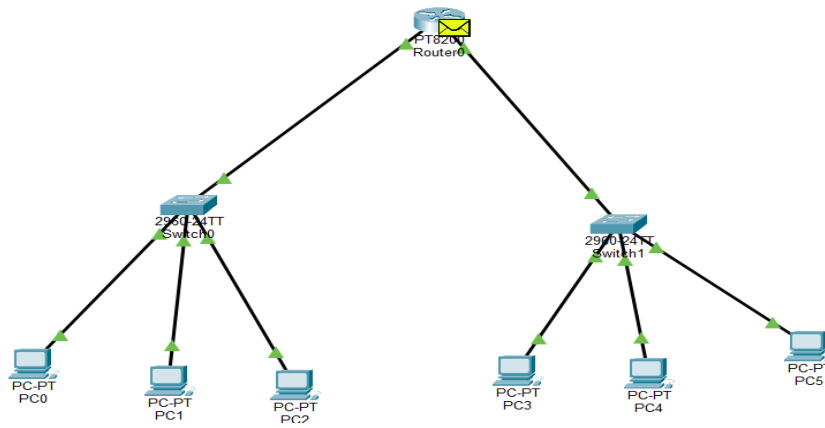
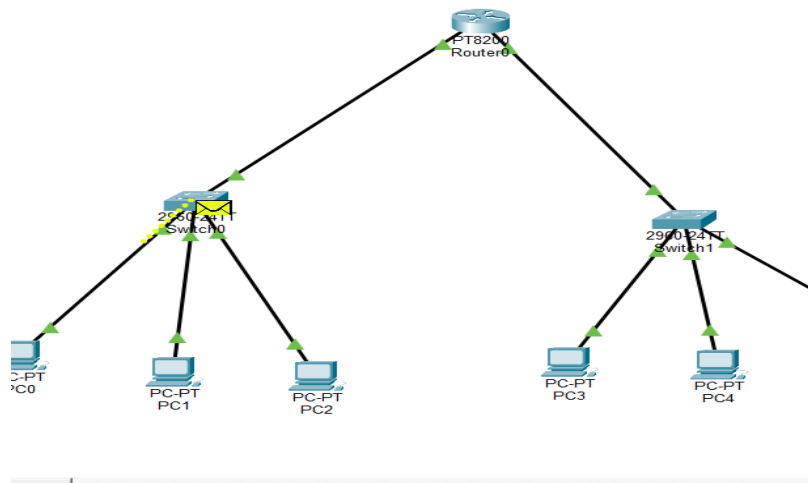
```

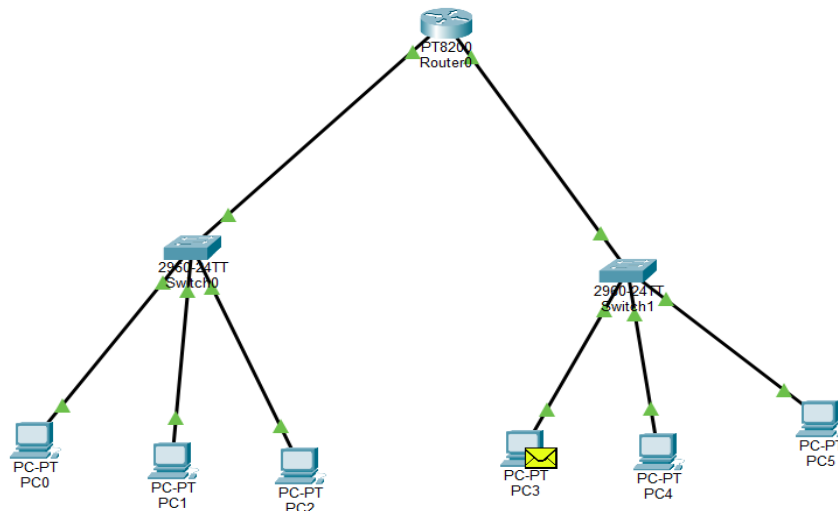
C:\>ping 10.10.10.2
Pinging 10.10.10.2 with 32 bytes of data:
Request timed out.
Reply from 10.10.10.2: bytes=32 time=1ms TTL=127
Reply from 10.10.10.2: bytes=32 time<1ms TTL=127
Reply from 10.10.10.2: bytes=32 time=1ms TTL=127

Ping statistics for 10.10.10.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>|

```

Output:





5. Repeater:

Working Method:

- Finally, we place a repeater between distant devices in the network.
- We connect devices on either side using straight-through cables.
- We then send data packets and observe that the repeater regenerates and amplifies the signal, extending the network range.
- This ensures that long-distance communication is maintained without data loss, showing the repeater's importance in larger network setups.

Ping from PC0 to PC5:

```
C:\>ping 192.168.1.7

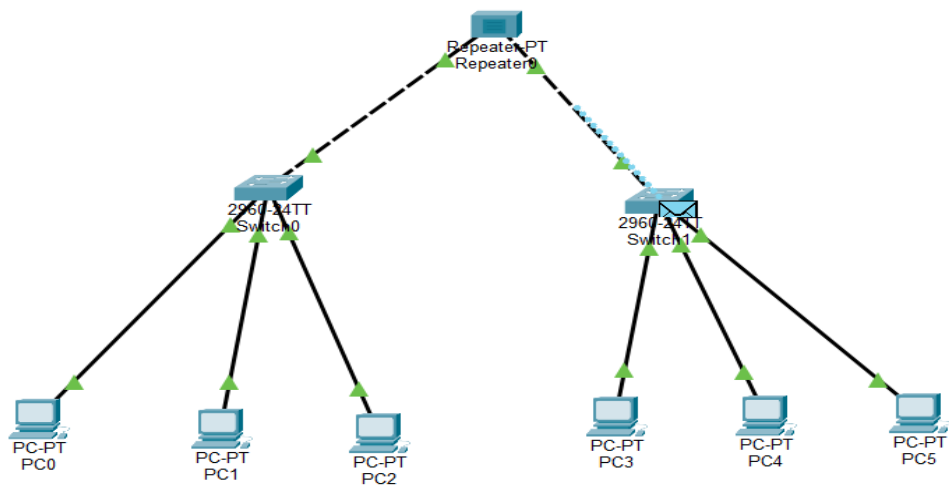
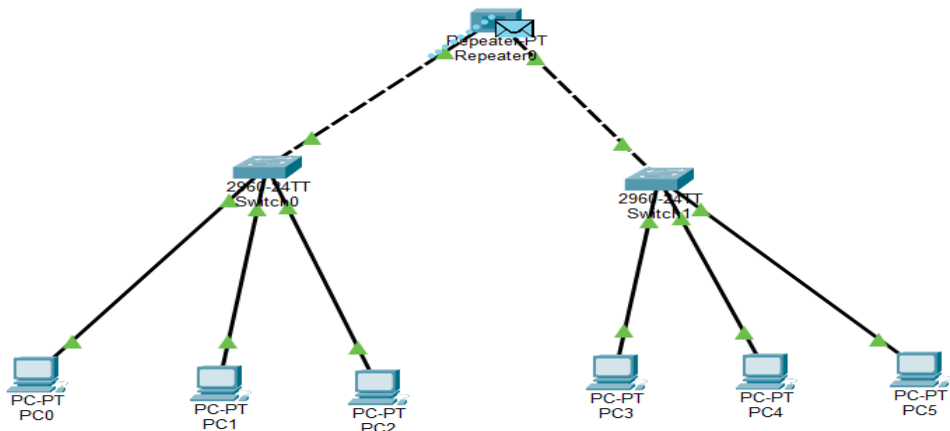
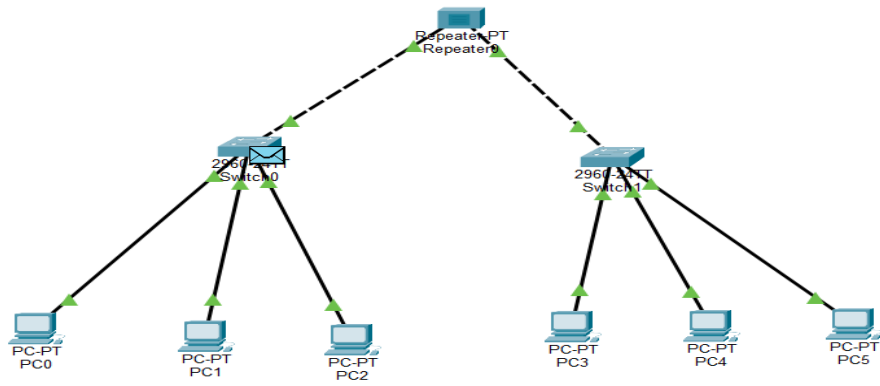
Pinging 192.168.1.7 with 32 bytes of data:

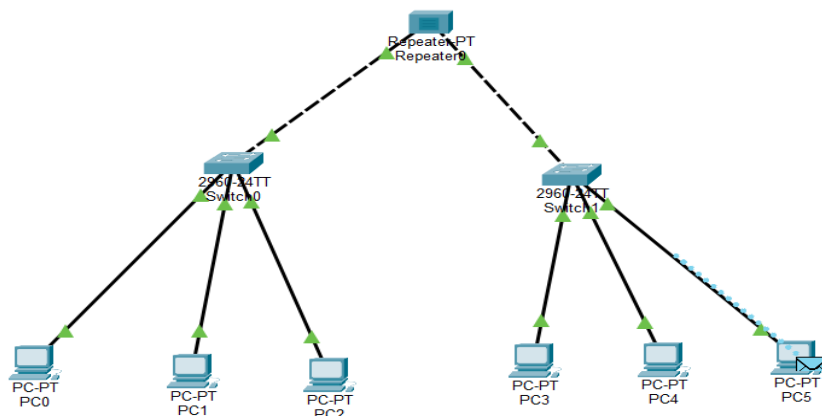
Reply from 192.168.1.7: bytes=32 time=3ms TTL=128
Reply from 192.168.1.7: bytes=32 time=1ms TTL=128
Reply from 192.168.1.7: bytes=32 time=7ms TTL=128
Reply from 192.168.1.7: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 7ms, Average = 3ms

C:\>
```

Output:





❖ Discussion and Conclusion:

In this lab, the simulation of basic network devices using Cisco Packet Tracer provided a clear understanding of how data is transmitted and managed within a computer network. By observing the behavior of hubs, it was evident that broadcasting data to all devices causes congestion and collisions. Switches and bridges demonstrated improved efficiency by filtering traffic using MAC addresses and reducing unnecessary data flow. Routers showed how different networks are interconnected using IP addresses and routing decisions. The repeater highlighted the importance of signal regeneration for long-distance communication. Cisco Packet Tracer proved to be an effective tool for visualizing packet flow and device functionality. Overall, this lab strengthened practical knowledge of network devices and their OSI layers. It also helped in comparing device performance and understanding real-world networking concepts.

